

Web Design Scraping: Enabling Factors, Opportunities and Research Directions

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Abstract—The number of online users accessing websites and consuming online services via their desktop and mobile browsers is on a constant rise. This paper coins and discusses the concept of web design scraping, which primarily promotes the idea of extracting, understanding, and modeling website components and characteristics and thereby inferring the meaning of the web design. Moreover, web design scraping emphasizes the role of state-of-the-art computing technologies and approaches, especially those powered by machine learning, in consolidating the understanding and intelligent construction and customization of website interfaces. Moreover, this paper discusses several enabling factors for web design scraping and recommends four research directions. The most notable research trends focus on the prediction of user satisfaction towards personalized web designs and the smart revamping of websites automatically by applying machine learning techniques that learn from the interaction history, preferences, and habits of online users.

Keywords—web design, scraping, intelligent user interface, artificial intelligence, machine learning, perceived usability, automation, user satisfaction

I. INTRODUCTION

The number of website users is on the rise, exceeding 4.54 billion active web viewers worldwide [1]. Moreover, there are approximately 1.72 billion websites accessible through desktops and smartphone devices [2]. However, research studies have reported that, on average, users abandon websites within the first 3 seconds due to diverse reasons, including the complexity and slow loading times of sites [3]. Therefore, developers must produce efficient yet compelling web designs that accommodate the taste of their users and keep up with the technological changes. Furthermore, current websites would need to be responsive to satisfy the enormous variability enforced by modern smartphones concerning screen size, resolution, and supported technologies. In user-centered software engineering, user testing is a vital ongoing activity to ensure that the software artifacts produced conform to the user needs and requirements, thus ensuring the highest levels of software acceptance and adoption by the end-users. However, conducting validation testing with real users is a sophisticated and laborious process that requires significant expertise, resources, and time.

Automation is the next big thing. The fast-technological pace, especially in the field of artificial intelligence (AI) and machine learning (ML), enabled the computerization of many human activities and business processes, spanning different areas from email spam detection to autonomous car driving. In the context of web design, the advantages offered by AI and ML can be exploited to automate both the creation of websites

and the evaluation of their quality in a smart way to reduce the burden and cost regarding software validation and maintenance [4]. Luckily, these advancements coincided with a similar technological shift in the field of human computer interaction (HCI) [5].

The benefits of integrating human computer interaction theories and artificial intelligence techniques are well recognized [6]. For example, it leads to increased performance, reduced costs, and higher user acceptance of the produced software. To realize an effective integration between the two fields, the right balance ought to be struck between human control and computer automation, which leads to the development of safe, reliable, and trustworthy systems. Despite the great promises anticipated, research efforts pushing towards AI-powered web design are quite limited in the literature. For instance, recent AI works concentrated on optimizing energy consumption and performance in mobile web designs [7], [8]. On the other hand, the proposed frameworks, techniques, and guidelines for generating intelligent and adaptive user interfaces are relatively generic [9] and do not apply well to the domain of web design.

In this paper, we coin and discuss an important concept, namely web design scraping. In simple terms, scraping web design refers to the idea of identifying, extracting, and understanding the visual properties of the elements that compose particular web designs to learn from the existing practices and predict efficient web models. Deducing the meaning of web design is expected to achieve various goals, such as the detection of design problems before user testing and the provision of recommendations to enhance the fitness between the websites and its users, among others.

This paper stipulates a set of research contributions that will encourage HCI and AI researchers and practitioners to pursue further research and advancements in the area of intelligent web design. The contributions are listed as follows:

- Introduce the overarching concept of web design scraping;
- Highlight the critical enablers of web design scraping, comprising modern web development technologies, image processing and machine learning approaches that may be employed to help in the comprehension and prediction of intelligent web designs;
- Propose several future research directions for achieving intelligent web design, such as the need to formalize the understanding of user interface components so to enable design automation.

In the next section, the paper lays the foundations of web design and web analysis tools with a focus on their shortcomings. Section III presents the concept of web design scraping. Section IV reminds of the most prominent factors that may boost the intelligent generation of web designs to fit the ever-evolving requirements and tastes of web users. Section V suggests four broad research directions for smart web design, while section VI discusses some of the current challenges against this vision.

II. PREVIOUS WORKS

A. Foundational Concepts of Web Design

Primarily web design focuses on designing, generating, and displaying content, in its varying formats, as websites over the Internet [10]. Today, website design is accessible through desktop browsers and mobile browsers, producing both static and dynamic websites. The process of designing websites involves crafting the visual appearance by integrating several elements, such as the layout, navigation, color, etc. [11]. Nonetheless, producing efficient and aesthetically pleasing web interfaces by designers and developers is no easy task.

The quality of a website directly impacts its usage and overall acceptability by web users [12], [13]. However, web quality is a broad concept that encompasses several multifaceted constructs, including content quality, branding, accessibility, social presence, information architecture, mobile communication, and usability, among other indicators [14]. Reference [15] defined four main characteristics of web applications' quality, specifically reliability, functionality, content, and usability. Other findings reported additional quality dimensions, such as personalization, navigation, appearance, multimedia, and design [16].

Generally, there is a consensus that the perceived web quality by users is influenced by the elements of web design [16], [17], [17]. Several factors are alleged to arouse user emotions and satisfaction concerning web design. These factors include color, animations, use of media, interactivity, and fonts attributes [19]. However, reference [20] showed that the typography, color, content quality, interactivity, and navigation are essential design factors for instilling user satisfaction and loyalty in websites. Moreover, reference [21] provided a more exhaustive comparison of the design variables, including visual design, color, typography, interactivity, and navigation, which could influence user attitudes and adoption towards the use of a particular website.

Moreover, web design is deemed as an essential aspect of the user experience design [11]. The user experience encompasses three primary constructs, namely, the overall look, the feel, and the perceived usability [11]. The overall look aims to create the surface or visual design of the system to enhance credibility and trust among the users. The feel seeks to facilitate and improve the interaction experience between the user and website to ensure enjoyment. Finally, the perceived usability is a user interface quality that is concerned with making the system efficient to use and reducing the usability issues.

The emergence of modern web technologies and frameworks has undeniably led to the creation of dynamic and interactive websites that cater to various types of interaction styles and devices. Nonetheless, the efforts to understand the composition and semantics of websites automatically are lagging. Timid works include the research of [22] who parsed

semi-structured documents (e.g., HTML documents) to extract content from news websites. Moreover, mastering web coding languages and frameworks does not necessarily produce winning websites. Essentially, part of the web developers' job is to implement the appearance of a website, including its structure, navigation, use of color, fonts, and media. However, the developers often lack the necessary understanding and knowledge of interaction design and psychology of their users to be able to create web designs that enrich the user experience and reduce the frustration caused by hidden interaction issues.

B. User Testing as an Integral Part of Effective Software Engineering

Software engineering contributes various development models that can be adopted and followed to produce software [23]. Although these models differ in the number of processes they encompass, their overall philosophy and complexity, they agree on the need to test the prototypes with users to capture and incorporate their views and comments in an iterative manner. In software engineering, this is often referred to as software validation. The user-centered design of software, although a useful design methodology to adopt to ensure the user interface designs meet user expectations, it adds cost and time to the development process. It also makes it challenging to satisfy various user profiles. Automating the design of user interfaces and acceptance testing brings about several benefits; for example, it expedites the testing process and allows developers to produce several parts of the design as well as reducing human bias and errors.

C. Efforts to Integrate Human Computer Interaction and Artificial Intelligence

Glancing over the legacy of HCI and AI expansion, one may recognize that their seemingly independent interests and priorities have shifted lately to be closer than ever. HCI aims to develop practical engineering methods to satisfy users when they interact with software applications [24]. In contrast, AI endeavors to design artificial agents that are competitors to human awareness [25]. Both HCI and AI strategies are still facing different types of challenges, leaving human desires not entirely satisfied. For instance, many AI researchers have not paid enough attention to human inputs while creating their AI models. Rather, they have concentrated on optimizing the AI models and algorithms [26]. For example, they have worked on the performance and error metrics (e.g., precision and recall) to evaluate their black-box models. Therefore, involving HCI practitioners in the development process of AI models is critical, particularly with the intelligent models that have an impact on human lives. HCI theories and models can be further exploited to assess important human metrics, such as fairness, truthfulness, understandability, interpretability, justifiability, security, etc. [26], [27].

Advancing HCI techniques using AI, on the other hand, can be usually encountered in reasoning approaches with expert systems, natural language processing (NLP), knowledge representation, and data mining [28]. This type of integration has given rise to the so-called Human-Centered AI (HCI) that aims to bridge the technological and humanistic perspectives for unifying the common concerns of human beings. For example, reference [29] proposed a deep learning framework for integrating visualization and analytics principles in a human-centered interface. Moreover, reference [30] presented a detailed guideline for the human-AI interaction design process.

D. Web Analysis and Evaluation Tools

There exist a plethora of web analysis tools devoted to evaluating the quality of websites from different perspectives [31]. As discussed earlier, quality may be viewed and assessed from several angles. Quality may be considered as a function of performance that optimizes the loading speed of the site and the time it takes to become fully interactive to online viewers. Amongst the popular performance analysis tools is Google Page Insights, which is an online performance tool that scores a given website on several metrics, including first contentful paint and time to interactive, and suggests remedies based on the detected issues. The quality check can be regarded as a function of the correctness and validity of the code used [31].

Other analysis tools, e.g., VisAssert [32], test the accessibility of websites to people with special needs against best practices accessibility guidelines. Other examples include Site Analyzer and Qualidator tool, which consider other aspects such as search engine optimization, content, design, and usability [31]. Other quality features analyzed include mobile-friendliness and security of websites. But evidently, there is a lack of tools that analyze web quality from a design perspective.

III. FROM CONTENT CRAWLING TO WEB DESIGN SCRAPING

Web scraping is not a novel concept, for it has been around for about two decades now. Harvesting or scraping the web refers to the extraction of text, automatically, from websites and storing it into a structured database for subsequent analysis. In other words, web scraping grabs and copies the content from websites [33]. The techniques used to achieve web scraping range from HTML parsing, HTTP programming, DOM parsing, vertical aggregation, and computer vision. We go beyond normal content scraping to user interface scraping of web designs.

In simple terms, we define ‘web design scraping’ as the process of extracting the visual properties of websites using suitable extraction and parsing techniques to construct an understanding of the web design that could be stored within appropriate models, as depicted in Fig. 1. Web design scraping goes beyond web content crawling to user interface design comprehension and modeling automatically. So, what makes scraping the web design a plausible and worthwhile research goal to pursue? There are various qualifying motivations to attempt to understand the visual interface automatically. Firstly, the task of mining web design elements enables the creation of models of design features (i.e., scraping design models as in Fig. 1) that can explain the perception of users towards websites. Secondly, it enables and clarifies the prediction of user judgment towards specific aspects of web design. Thirdly, capitalizing on the UI design knowledge would make the goal of AI-based design automation a reality and thus reduce the implicated cost and resources required to develop and integrate websites. Fourthly, scraping allows the design to adapt to the evolving user patterns and respond quickly to the technological changes.

IV. THE INGREDIENTS ENABLING THE PREDICTION OF USER SATISFACTION AND ACCEPTANCE OF WEB DESIGN

Several research studies have attempted to model the users during their interaction with websites and anticipate their behaviors and impressions at the end of the interaction process, e.g., [34]. However, predicting user judgment and

satisfaction with websites is a complex goal due to manifold reasons.

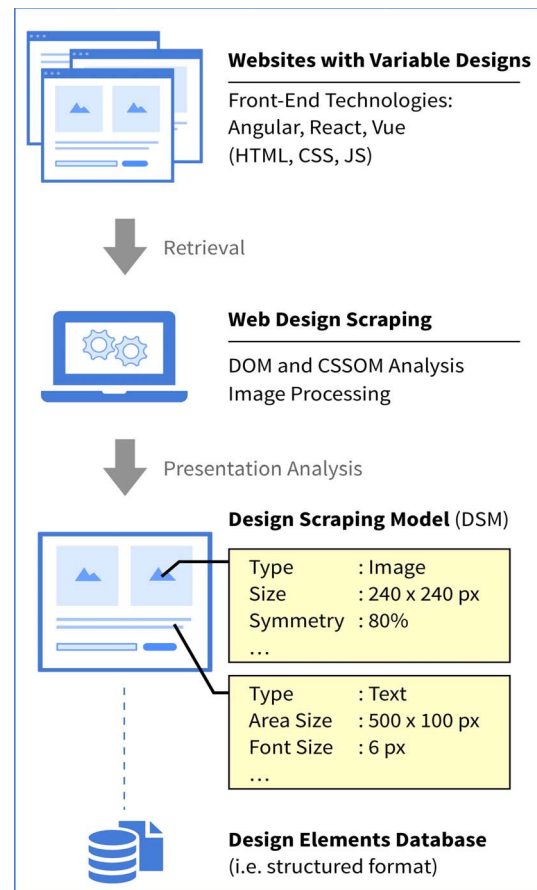


Fig. 1. Our View of Web Design Scraping

Firstly, evaluating web designs is a natural subjective process that depends on the profile, characteristics, and past experiences of users. Such evaluation often varies from one person to another. Secondly, user needs and tastes usually differ between groups, which makes the UI designs unsatisfiable to all of them, thus requiring customization. Thirdly, the evaluation consists of perceived factors (e.g., design originality) that may be hard to calculate and predict computationally. Fourthly, the interaction between the design factors is still not well understood yet and requires further research studies. However, here we review some recent trends and factors that are believed to enhance the prospect of modeling and anticipating user acceptance and satisfaction of websites.

A. The Emergence of Modern Web Technologies and Frameworks

The information proliferation is an artifact of the revolutionarily advances that took place in web technologies together with the hardware breakthroughs. Modern front-end web development frameworks rely heavily on three cornerstone technologies, namely Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript to deliver interactive and responsive websites [35]. Previously, web developers had used Java and Flash plugins to deliver animated content, but these technologies have become obsolete now [36].

Nowadays, Internet users are increasingly accessing websites through various devices with different screen sizes,

owing to the adaptive and responsive web design strategies [10]. These techniques adopt the latest HTML, CSS, and JavaScript versions to develop the front-end (i.e. HTML5, CSS3, and JavaScript 1.8.5 respectively). HTML documents are text files used for marking up the website elements, while CSS rules are used for defining how these elements are structured and how the web pages are visually represented to the user [35]. JavaScript, on the other hand, enables the implementation of user events and interaction. The availability of front-end development libraries and frameworks made client-side development faster, easier, and more organized. Example JavaScript frameworks include ReactJS, AngularJS, and VueJS [37].

A web browser makes use of HTML and CSS files to construct the so-called Document Object Model (DOM) and CSS Object Model (CSSOM) trees separately [7]. Next, a render tree is generated from the DOM and CSSOM, as shown in Fig. 2. The render tree contains the website nodes, including the styling information and text content, which later become the basis for the web browser to paint on the web viewport [7]. It is reasonably easy to construct a web design scraping model that encompasses the visual properties and design features of websites from the render tree. While some design attributes, e.g., the position and font type, can be inferred easily using dedicated JavaScript APIs [37], other more complicated attributes, e.g., complexity and symmetry, may require the use of sophisticated techniques, e.g., image processing.

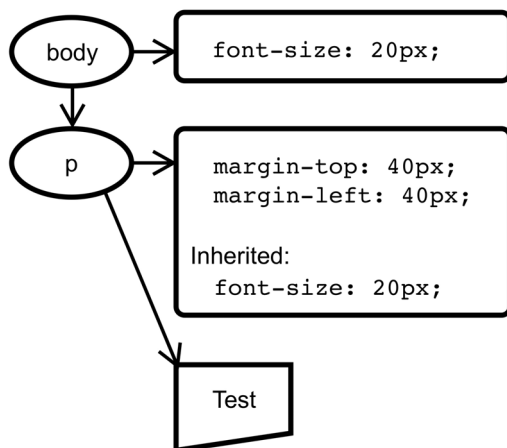


Fig. 2. A Render Tree Generated from the DOM and CSSOM Models

B. Image Processing Techniques

Image processing techniques, in computer vision, take images as input and apply analysis algorithms to either extract certain desired information (e.g., objects and features) or manipulate the input images to achieve some desired outputs. Such techniques have enjoyed vast applications in several fields, for example, medical science, military, environment, and robotics. These techniques range from basic image compression, restoration, editing, transformation, classification, to complicated AI-driven feature extraction and pattern recognition [38].

The surface of a website is the principal means to communicate information to the web viewers and it needs to be visually appealing [39]. The abundance of image

processing techniques empowers designers to study the visual components of a website more accurately, especially the perceived factors that cannot be examined by the mere analysis of the website files. Reference [40] showed that it is possible to compute and improve the transmission of web information by focusing on typography, images, color, and knowledge structure. Algorithms already exist, with a reasonable level of success, to detect complicated perceived visual concepts, such as symmetry and complexity, within mobile screens [41]. These visual attributes are well known to influence the perceived aesthetics of web design.

C. Artificial Intelligence and Self Learning Behavior

Building intelligent web design models using ML is no different than developing other tools that utilize AI techniques. ML aims to produce mathematical models that support understanding (e.g., by clustering) and prediction (e.g., by classification) through the analysis of the available data in three subsequent steps: (1) feature engineering, (2) model selection, and (3) training, and presenting the results [25]. Let us instill these three steps within the framework of developing intelligent web scraping models.

Feature engineering refers to several tasks focusing on data preparation, including feature selection, data cleaning, normalization, and encoding. With scraping extensive messy web data, one can imagine the difficulty of this step as having sparse data often leads to getting inaccurate predictions with poor performance, and vice-versa. To a large extent, this step can be tackled by (1) information extraction and regular expression techniques using popular NLP tools [42], or (2) text mining by deep statistical learning. It is worth mentioning that one of the most promising textual analysis techniques to date relies on the recurrent neural networks (RNN) [43]. An available tool that beautifully engineers scrapped web data and make it ready for ML models to use is Diffbot.

It is quite essential to select and combine the appropriate ML models to understand and come up with reasonable predictions about web design elements. This is because web design elements have different data types and visual attributes. For instance, Convolutional Neural Networks (CNN) is the best encouraging model for image/video and audio analysis [43] as it can handle, e.g., displayed HTML shape elements like an image of a textbox. However, RNN may be used to analyze the textual DOM elements. With deep learning capabilities by neural networks, many computer vision tools have emerged to help transfer handwritten designs into complete web designs with few clicks, e.g., Microsoft-Sketch2Code¹.

Concerning the third step, training and presenting the results can be performed adaptively using an online efficient reinforcement learning approach [44] or discontinuously over time using regular (un) supervised learning methods. Adaptive learning typically does not need to start with a large dataset; instead, it understands user preferences by incrementally collecting site and user metadata (e.g., publicly available in social media platforms). In return, it gives users personalized recommendations, such as suitable UI layouts and themes. While this is also possible to obtain using the second approach, the generation of a complete website from scratch requires training large datasets that should include thousands

¹ Microsoft Sketch2Code uses AI to convert hand-written drawings to working HTML prototypes. <https://www.microsoft.com/en-us/ai/ai-lab-sketch2code>

of templates mapped with different user needs and goals. Wix², for example, is an AI tool that implements the second approach for designing and generating a complete website in a matter of seconds.

D. Research of Web Design Factors and Usability Guidelines

Human Computer Interaction knowledge, formalized as paradigms, theories, models, and guidelines, will be a vital component to anticipate usability problems and user frustration before they occur [45], [46]. Moreover, the studies, e.g. [3], [20], [39], which were conducted concerning web design produced a wealth of suggestions and principles that will help maximize efficiency, accessibility, safety, and usability of intelligent web designs. However, further research is required to organize, clarify, and validate these design specific recommendations [45].

V. CONCLUSIONS AND FUTURE RESEARCH DIRECTIONS

This paper positions and coins a new term, namely “web design scraping”, that is fundamentally different from the regular web scraping. It goes beyond harvesting and extracting web content to analyzing websites of varying complexity with a focus on gathering information about their design aspects and thus forming a semantic understanding of the design. Web design scraping promises to empower site developers and owners to enhance the overall perceived usability and acceptability of their websites through research into the four areas discussed below.

A. Area One: Understanding the meaning of web design

The first step towards realizing the goals of this paper is to utilize the available web technologies and techniques to establish a full understanding of the design elements composing websites on various types of devices, including mobile devices, which have become the mainstream technology in our daily life. Therefore, understanding web design has to span both desktop and mobile user interfaces. Front ends of websites that are built entirely using HTML and CSS can be easily parsed and inspected to infer the features, e.g., typography and navigation, that make up the appearance of websites. The parsing could be synthesized with other smart computer vision methods to recognize and extract more complicated features about visual perception, such as balance and symmetry.

B. Area Two: Inferring web design flaws

Forming a deep understanding of the web design and its semantics enables the inference of possible design issues by subjecting the web development and prototyping practices to the research principles and guidelines. The general goal here is to inform the website designers and owners of the design discrepancies early before real user testing takes place or market release so that the detected issues can be fixed as part of the user-centered design methodology. Developers are usually tempted to jump into coding without observing the best design practices, leading to serious design problems and additional maintenance costs. The first step here is to come up with a language to express and formalize the best practices and guidelines into formal assertions focusing on usability, aesthetics, accessibility, and web design in general. Such formalizations could be categorized according to their use and purpose (e.g., typography, device) and used to evaluate the

websites. Future research efforts should also concentrate on predicting usability issues before they happen to ensure optimal user experience and repeated visits. Such predictions will help the developers, who are usually not design experts by profession, to discover the user interface issues early and react to them.

C. Area Three: Suggesting design improvements

It is always beneficial to suggest practical solutions to overcome the design issues highlighted during the testing phase of the websites. Such recommendations ought to be founded and based on heuristics, rules, and theories extracted from the HCI research literature. The recommendations should cover all factors that are predicted to impact the perceived usability, aesthetics, and satisfaction of the web viewers. This strand would need to emphasize how it is possible to fulfill the guidelines using the modern web frameworks and implementation languages. Design improvements would need to cover both desktop and mobile sites.

D. Area Four: Intelligent refactoring of web designs

The previous areas of research would accumulate the necessary knowledge and pave the way for re-designing interactive, personalized, and accessible websites intelligently. Creative re-design would consider the interaction patterns and behaviors of web users and generate new websites to match their demographics, tastes, and needs. This could be achieved by profiling the users and their preferences as well as their historic web visits. Currently, popular websites keep track of historical data of their web visitors and adjust their recommendations of products and services based on these data. The same may be achieved with respect to rendering web representations. The regeneration process would require formalizing the visual appearance, such as the layout, of the website to be able to adapt the web appearance to multiple screen sizes and devices automatically.

Moreover, it would be beneficial to generate multiple design solutions and compare them through user testing before committing to the final web design. The smart generation of customizable designs will enable design practitioners to satisfy the requirements of people with special needs and the elderly. For instance, specific interaction styles and modes could be provided based on user abilities and viewing patterns.

VI. BARRIERS AND CHALLENGES

As much as the concept of web design scraping might sound appealing, several challenges need to be highlighted and discussed in this paper. The significant variability and mix of web implementation languages and frameworks complicate the process of fully understanding the web designs. This is because different scraping models have to be developed and integrated to deal with various web programming languages. Moreover, the delivery and accessibility of websites through versatile devices, including portable computing devices, add more burden to the semantic understanding of design. For instance, the knowledge extracted about a specific webpage displayed on a regular computer screen might not necessarily be the same on a smartphone. Today, portable and handheld devices have their particular characteristics and differ significantly in screen sizes, software, and hardware they offer and support. Creating web design scraping tools that

² Wix is an AI website builder, <https://www.wix.com>

accommodate the changing nature of mobile devices is one of the significant challenges. Restructuring web designs, although feasible, may be limited by the fact that specific creative web designs could be challenging to generate, and human intervention would always be needed to inject an element of imagination and appeal. Finally, design recommendations and aesthetics principles are not universal laws of usability and beauty.

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